

Sex- and Menopause-Related Differences in Immune and Gastrointestinal Symptom Architecture in ME/CFS: Evidence from Factor Analysis and Structural Equation Modeling

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Research Article

Keywords: ME/CFS, symptom clusters, sex differences, neuroimmune dysfunction, gastrointestinal symptoms, menopause, dysbiosis, structural equation modeling.

Posted Date: June 30th, 2026

DOI: <https://doi.org/10.21203/rs.3.rs-9825975/v1>

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Title:

Sex- and Menopause-Related Differences in Immune and Gastrointestinal Symptom Architecture in ME/CFS: Evidence from Factor Analysis and Structural Equation Modeling

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Abstract

Background:

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a multisystem disorder characterized by neuroimmune, autonomic, and gastrointestinal dysfunction. Previous studies identified coherent symptom domains involving *Brain*, *Autonomic*, *Gut*, and *Immune* manifestations and demonstrated pronounced sex-specific differences in neurocognitive–sensory and autonomic symptom organization. However, whether immune-related and gastrointestinal symptoms form distinct or integrated latent structures in women and men with ME/CFS remains unclear. In addition, the potential influence of menopausal status on these symptom domains has not been systematically investigated.

Methods:

Data from 748 adults with medically diagnosed ME/CFS (608 women, 137 men) from the APAV-ME/CFS study were analyzed. Symptoms were coded dichotomously. Sex-stratified analyses included cross-tabulations, Cramér's V, tetrachoric correlations, logistic and linear regression models, exploratory factor analysis, and structural equation modeling (SEM). Additional subgroup analyses compared pre- and postmenopausal women. Model robustness was evaluated using a stratified training dataset.

Results:

In women, *Immune* (flu-like symptoms, susceptibility to infections) and *Gut* (gastrointestinal complaints, food intolerances) symptoms formed two distinct but related clusters, characterized by moderate within-cluster correlations (tetrachoric $\rho = .47\text{--}.60$) and weaker cross-cluster associations ($\rho = .30\text{--}.35$). Exploratory factor analysis supported a two-factor solution, and SEM demonstrated superior fit for a two-factor model (CFI = .913; RMSEA = .119).

In men, all four symptoms loaded onto a single integrated *Gut/Immune* factor, with consistently high intercorrelations across domains ($\rho = .43\text{--}.68$). SEM confirmed excellent fit for a one-factor model (CFI = .981; RMSEA = .074). Susceptibility to infections emerged as the dominant predictor in regression analyses.

Among women, menopausal status selectively affected immune-related symptoms. Premenopausal women reported flu-like symptoms more frequently than postmenopausal women, whereas gastrointestinal symptoms and food intolerances remained stable across groups. These findings suggest differential hormonal sensitivity of immune versus gastrointestinal symptom trajectories.

Conclusions:

Immune and gastrointestinal symptoms in ME/CFS exhibit sex-specific latent structures. Women demonstrate two partially separable but related symptom domains, whereas men show

a more integrated immune–gastrointestinal architecture. Furthermore, menopausal status appears to selectively modulate immune-related symptom expression while leaving gastrointestinal symptom patterns comparatively stable. Together, these findings support the concept of sex-dependent neuroimmune–autonomic mechanisms in ME/CFS and are consistent with a dynamic, hormonally modulated immune component alongside a more persistent gut-related process. The results highlight the importance of sex- and hormone-sensitive approaches to phenotyping, mechanistic research, and therapeutic stratification in ME/CFS.

Keywords:

ME/CFS; symptom clusters; sex differences; neuroimmune dysfunction; gastrointestinal symptoms; menopause; dysbiosis; structural equation modeling.

Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating multisystem disease [1] characterized by post-exertional malaise (PEM) [2], neurocognitive impairment [3], autonomic dysfunction [4], immunological abnormalities [5, 6], and a broad spectrum of fluctuating somatic symptoms [7]. The disorder is typically triggered by infectious events and involves disturbances across multiple regulatory systems—including the nervous, endocrine, and immune systems—as well as impaired cellular energy metabolism and chronic low-grade inflammation, potentially contributing to disturbed physiological homeostasis [8, 9]. These findings are consistent with recent multi-omics and immunometabolic research reporting evidence for exaggerated innate immune responses, chronic inflammation, metabolic dysfunction, and gut dysbiosis in ME/CFS patients.

Despite the multisystemic nature of ME/CFS, relatively little is known about how symptoms co-occur, cluster, or interact across biological domains. In a previous population-based study (Study 1), we showed that many symptoms appear in parallel across the disease course and that gastrointestinal and immunological symptoms frequently emerge early, particularly in women [8]. Building on these observations, Study 2 statistically validated four hypothesis-driven symptom clusters—*Brain*, *Autonomic*, *Gut*, and *Immune*—using exploratory and confirmatory factor analyses and structural equation modeling (SEM). These clusters demonstrated strong internal coherence and distinctiveness, supporting the hypothesis that ME/CFS symptom architecture may reflect underlying neuroimmune and autonomic processes [9].

Study 3 extended this work by examining sex-specific differences in the *Brain/Brain2*² and *Autonomic* clusters³. The results revealed marked sex-related divergence: women exhibited more homogeneous neurocognitive-sensory and autonomic clusters, whereas men showed more heterogeneous patterns suggesting vascular–metabolic and sympathetic involvement [10]. These findings align with established sex differences in immune function, where women typically show stronger adaptive immune responses and higher autoimmune activation, while men may exhibit weaker humoral immunity and greater susceptibility to certain infections [11].

Gastrointestinal symptoms are particularly relevant in ME/CFS. Multiple studies have documented alterations in the gut microbiota, reduced microbial diversity, and shifts toward pro-inflammatory taxa, suggesting a role for dysbiosis in disease pathophysiology. Recent

1 The *Brain* cluster consisted of the symptoms brain fog, headaches, sleep disorders, visual disturbances, and sensory hypersensitivity. The *Brain2* cluster additionally includes the symptom of temperature incompatibility.

3 The *Autonomic* cluster includes symptoms such as cardiovascular problems, breathing problems, gastrointestinal complaints, and temperature incompatibility.

reviews highlight that dysbiosis may contribute to intestinal barrier dysfunction (“Leaky Gut”), microbial translocation, oxidative stress, and systemic immune activation, potentially exacerbating ME/CFS symptoms [12, 13]. Although causality remains unproven, these mechanisms are biologically plausible and consistent with broader evidence linking gut dysbiosis to neuroinflammation and blood–brain barrier alterations [14]. A 2024 review further emphasizes that gastrointestinal microbiota changes may influence immunological and inflammatory pathways relevant to ME/CFS, even though current evidence is largely associative [15].

Taken together, these findings suggest that immune and gastrointestinal symptoms may form interconnected clusters, potentially mediated by gut–immune–brain interactions. However, whether these clusters manifest similarly in women and men remains unclear. Preliminary evidence from Study 2 and Study 3 suggests that women may exhibit distinct *Immune* and *Gut* clusters, whereas in men these symptoms may converge into a single integrated cluster. Such sex-specific differences would be consistent with known biological patterns: women generally exhibit stronger adaptive immune responses and may show greater vulnerability to immune–gut interactions, whereas men may display stronger sympathetic and vascular–metabolic involvement as well as greater susceptibility to certain infections [11, 16–18].

The present study (Study 4) therefore aims to: (1) determine whether *Immune* and *Gut* symptoms form two distinct clusters in women but a single integrated cluster (*Combined Gut/Immune*) in men; (2) evaluate the statistical coherence, distinctiveness, and latent structure of these clusters using bivariate associations, factor analysis, and SEM; and (3) situate these findings within a broader mechanistic framework linking immune–gut interactions to autonomic and neurocognitive symptom domains.

By integrating statistical modeling with emerging biological insights, this study seeks to clarify sex-specific symptom architectures in ME/CFS and contribute to a hypothesis-generating model that reflects divergent neuroimmune–autonomic pathways in women and men.

Methods

Study Design and Data Source

This study (Study 4) is part of a multi-phase research program investigating symptom architecture in myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS). It builds on findings from: (1) a population-based descriptive study of symptom frequencies (Study 1), (2) a factor-analytic validation of mechanistic symptom clusters (Study 2), and (3) an analysis of sex-specific differences in neurocognitive-sensory and autonomic clusters (Study 3).

Study 4 focuses on the immune-related and gastrointestinal symptom domains. All analyses were conducted using data from the APAV-ME/CFS study, a predominantly qualitative investigation of the doctor–patient relationship that also collected quantitative symptom data.

Participants

A total of 1,238 individuals participated in the APAV-ME/CFS study. For the present analyses, we included only adult participants (>20 years) with a medically confirmed ME/CFS diagnosis ($n = 748$; 608 women, 137 men, 3 non-binary). Exclusion criteria were:

- age ≤ 20 years,
- absence of a medical ME/CFS diagnosis,
- relatives of ME/CFS patients,
- incomplete information on key demographic variables (sex, age, diagnosis).

The sex distribution ($\approx 81\%$ women) is consistent with epidemiological data reporting a female predominance of 70–80% in ME/CFS (Figure 1).

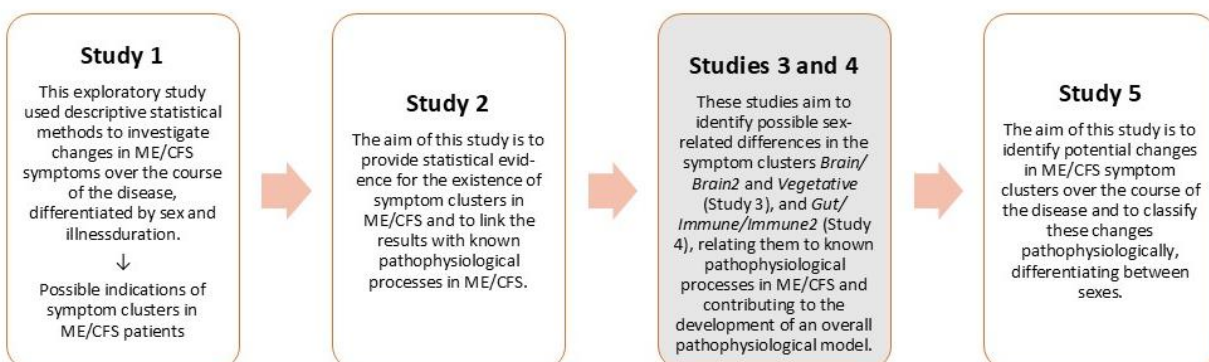


Figure 1 Chronological placement of study 4 within the sequence of studies.

Symptom Assessment and Coding

Participants reported whether each symptom had occurred regularly in recent months. All symptoms were coded as **dichotomous variables** (0 = absent, 1 = present). Dichotomization reduces variance and may attenuate factor loadings and SEM fit indices; however, global model fit indices in Studies 2 and 3 demonstrated that the latent structure remains robust despite this limitation.

The present study focused on four symptoms forming the hypothesized *Immune* and *Gut* clusters:

- *Immune:*
 - flu-like symptoms
 - increased susceptibility to infections
- *Gut:*
 - gastrointestinal complaints
 - food and other intolerances

For integration analyses, we also considered the previously validated *Autonomic* and *Brain/Brain2* clusters.

Statistical Analyses

All analyses were performed separately for women and men to examine sex-specific symptom structures. The latent variables identified in the present analyses represent statistical symptom structures rather than directly measured biological mechanisms.

1. Bivariate Associations

We first examined pairwise associations between symptoms using:

- cross-tabulations and Pearson's χ^2 tests,
- Cramér's V (effect size for dichotomous variables),
- Pearson correlations,
- tetrachoric correlations, which assume an underlying continuous latent distribution and are appropriate for binary indicators.

2. Logistic Regression

Binary logistic regression models were estimated to assess the predictive relationships between symptoms. Depending on the analysis, either flu-like symptoms or susceptibility to infections served as the dependent variable. Model fit was evaluated using pseudo- R^2 statistics.

3. Linear Regression

Composite symptom scores were calculated by summing dichotomous symptom indicators to approximate continuous variation in symptom burden. Scores represented the sum of present symptom indicators and were treated as approximately continuous variables. The *Immune* score consisted of two dichotomous items (flu-like symptoms, increased infection susceptibility), yielding a possible range of 0–2. The *Gut* score likewise comprised two dichotomous items (gastrointestinal complaints, food intolerances), also ranging from 0–2. The *Combined Gut/Immune* score represented the sum of all four indicators and therefore ranged from 0–4. No standardization or z-transformation was applied. All analyses were conducted using raw sum scores to preserve interpretability.

4. Factor Analysis

Exploratory factor analysis (EFA) with Varimax rotation was performed to evaluate the latent structure of the four symptoms. In women, we examined whether *Immune* and *Gut* symptoms separated into two distinct factors, whereas in men we tested a single-factor solution representing a *Combined Gut/Immune* construct. Given that each domain comprises only two indicators and the overall model includes four items, the factor analyses are considered exploratory and hypothesis-driven. Accordingly, we do not interpret the factor loadings in detail but use them to assess whether the observed symptom patterns align with the proposed latent structure.

5. Structural Equation Modeling (SEM)

Confirmatory factor analyses (CFA) and SEM were conducted to test the fit of one-factor vs. two-factor models. All models were estimated using maximum likelihood (ML) in Stata 12.0 because categorical SEM estimators (e.g., WLSMV) were not available in the software version used. Because the indicators were binary, ML estimation may yield biased standard errors and fit indices. However, tetrachoric correlations and cross-validation analyses suggested that the latent structure was sufficiently robust.

This limitation was addressed by:

- using tetrachoric correlations,
- cross-validating results with CFA convergence,
- replicating all analyses in a stratified training dataset.

Model fit was evaluated using:

- χ^2 statistics,
- Comparative Fit Index (CFI),
- Root Mean Square Error of Approximation (RMSEA),
- Standardized Root Mean Square Residual (SRMR).

However, ML can skew standard errors for dichotomous items, distort the RMSEA, and affect factor loadings. Results should therefore be interpreted cautiously, particularly the absolute fit indices. Furthermore, the male subgroup size was relatively small for latent variable modelling. Therefore, these findings should also be interpreted cautiously and considered preliminary.

6. Stratified Training Dataset

To assess robustness, all analyses were repeated in a stratified, randomized training dataset derived from the full sample. Stratification was based on sex and illness duration to ensure proportional representation and minimize sampling bias. Results were directionally consistent across datasets.

7. Software

All analyses were conducted using Stata 12.0. Graphical models were created using Microsoft PowerPoint.

Results

Sample Characteristics

Analyses were based on 748 adults with a medically confirmed ME/CFS diagnosis (608 women, 137 men, 3 non-binary). Because the non-binary group was too small for statistical modeling, results are reported for women and men only. Symptom frequencies for the *Immune* and *Gut* domains were consistent with previous studies: flu-like symptoms were reported by

76.8% of women and 73.0% of men; gastrointestinal complaints by 71.9% of women and 64.5% of men; food intolerances by 63.2% of women and 47.8% of men; and increased susceptibility to infections by 48.0% of women and 43.8% of men.

1. Bivariate Associations

In both sexes, within-domain associations were stronger than cross-domain associations. Associations tended to be stronger in men than in women (Tables 1 and 2).

Table 1 Bivariate associations within the immunological symptom group (*Immune*) by sex

Feature	Women (n=608)	Men (n=137)
Pearson χ^2 (1)	44.5853	22.4052
p-value	< .001	< .001
Cramér's V	.2708	.4044
Pearson correlation	.27 (p < .001)	.40 (p < .001)
Tetrachoric correlation	.47 (SE = .060, p < .001)	.681 (SE = .097, p < .001)
Logistic regression – Pseudo R ²	.0709	.1581
OR (95% CI)	4.08 (2.65–6.27), p < .001	10.50 (3.46–31.87), p < .001

Bivariate associations within the gastrointestinal symptom group (*Gut*) by sex

Feature	Women (n=608)	Men (n=137)
Pearson χ^2 (1)	94.5498	23.5576
p-value	< .001	< .001
Cramér's V	.3943	.4147
Pearson correlation	.39 (p < .001)	.41 (p < .001)
Tetrachoric correlation	.603 (SE = .050, p < .001)	.630 (SE = .097, p < .001)
Logistic regression – Pseudo R ²	.1288	.1387
OR (95% CI)	6.18 (4.20–9.09), p < .001	6.83 (3.01–15.49), p < .001

Cross-cluster associations (*Immune* ↔ *Gut*) in men were markedly stronger (tetrachoric p: .43-.60) than in women (tetrachoric p: .30 to .35).

2. Logistic Regression

Women

In women, both immune-related and gastrointestinal symptoms contributed to the prediction of flu-like symptoms. In men, susceptibility to infections emerged as the only significant predictor (Table 3).

Table 3 Logistic regression (Outcome: flu-like symptoms)

Feature	Women	Men
Pseudo R ²	.0924	.1764
Increased susceptibility to infections	OR 3.448 (SE: .7768), $p < .001$	OR 7.5766 (SE: 4.5197), $p = .001$
Gastrointestinal complaints	OR 1.6868 (SE: .3848), $p = .022$	OR 1.5833 (SE: .7459), $p = .329$
Food intolerances	OR 1.5278 (SE: .3389), $p = .056$	OR 1.6019 (SE: .7994), $p = .345$

3. Structural Equation Modeling (SEM)

To examine the dimensional structure of *Gut* and *Immune* symptoms, three structural equation models (SEMs) were estimated: two separate one-factor models for *Gut* and *Immune* symptoms, respectively, and one combined one-factor model including all four symptom indicators.

The first model represented a latent *Gut* factor indicated by gastrointestinal complaints and food intolerances. The factor loading for food intolerances was moderate ($\lambda = .4231$), while residual variance remained relatively high, suggesting a weaker association with the latent construct. The second model represented a latent *Immune* factor indicated by flu-like symptoms and susceptibility to infections. In this model, susceptibility to infections showed a comparatively strong factor loading ($\lambda = .74$) and lower residual variance, indicating a strong association with the latent immune-related construct. These models are just-identified ($df = 0$). Therefore, fit indices for the separate two-indicator models are not informative.

A third model specifying a single latent factor underlying all four symptom indicators demonstrated substantially poorer fit to the data, $\chi^2(2) = 23.78$, $p < .001$. Within this combined model, gastrointestinal symptoms loaded strongly on the latent factor ($\lambda = .91$ and $\lambda = 1.00$), whereas immune-related symptoms showed considerably weaker loadings ($\lambda = .54$ and $\lambda = .74$).

Taken together, these findings support the distinction between two symptom domains rather than a single common construct. Accordingly, gastrointestinal symptoms (gastrointestinal complaints and food intolerances) and immune-related symptoms (susceptibility to infections and flu-like symptoms) were treated as separate latent dimensions in subsequent analyses.

Overall, the findings suggest sex-specific differences in the organization of gastrointestinal and immune-related symptoms. In women, the data support a differentiated two-domain structure, whereas in men the symptom domains appear more integrated into a single latent construct (Tables 4 and 5). However, findings in men should be interpreted cautiously due to the smaller sample size.

Table 4 Factor loadings of the *Combined one-factor model (Gut and Immune)*

Women

Indicator	Loading (λ)		p
	Unstandardised	Standardised	
Susceptibility to infections	1.00 (fixed)	.39	—
Flu-like symptoms	.757	.35	< .001
Gastrointestinal complaints	1.36	.60	< .001
Food intolerances	1.51	.66	< .001

Men

Indicator	Loading (λ)		p
	Unstandardised	Standardised	
Susceptibility to infections	1.00 (fixed)	.68	—
Flu-like symptoms	.67	.51	< .001
Gastrointestinal complaints	.85	.59	< .001
Food intolerances	.88	.60	< .001

Table 5 Fit indices of the *Combined one-factor model (Gut and Immune)*

Fit index	Women	Men	Fit index	Women	Men
χ^2 (df)	19.315 (2)	3.501 (2)	TLI	.739	.944
p value	< .0001	.174	SRMR	.040	.028
RMSEA	.119	.074	CD	.598	.698
90% CI RMSEA	.075–.170	.000–.200	AIC	2988.532	690.806
CFI	.913	.981	BIC	3041.454	725.846

4. Communalities and uniquenesses

The analysis of communalities (h^2) and uniquenesses (u^2) revealed clear sex-specific differences in the underlying factor structure. Among women, the two immune-related indicators *Susceptibility to infections* and *Flu-like symptoms* showed very low communalities ($h^2 = .15$ and $.12$, respectively), indicating that these items were only weakly accounted for by a common latent factor. In contrast, the gastrointestinal indicators *Gastrointestinal complaints* and *Food intolerances* demonstrated moderate communalities ($h^2 = .36$ and $.44$), reflecting a substantially stronger association with a GI-related factor. This pattern supports the identification of a two-factor solution in women, comprising distinct immune and gastrointestinal dimensions.

In men, a more homogeneous pattern emerged: all four indicators showed moderate communalities ($h^2 = .26$ – $.46$), suggesting that each item contributed meaningfully to a single latent construct. Correspondingly, uniqueness values were in the mid-range ($u^2 = .54$ – $.74$), with no indicator standing out as poorly represented. This profile is consistent with a unified, overarching factor integrating both immune-related and gastrointestinal symptoms. Overall, the communality patterns strongly support the sex-specific model differences: a two-factor structure in women versus a one-factor structure in men (Table 6).

Table 6 Communalities (h^2) and uniquenesses (u^2)

Women			
Indicator	λ_{std}	h^2	u^2
Susceptibility to infections	0.39	0.15	0.85
Flu-like symptoms	0.35	0.12	0.88
Gastrointestinal complaints	0.60	0.36	0.64
Food intolerances	0.66	0.44	0.56
Men			
Indicator	λ_{std}	h^2	u^2
Susceptibility to infections	.68	0.46	0.54
Flu-like symptoms	.51	0.26	0.74
Gastrointestinal complaints	.59	0.35	0.65
Food intolerances	.60	0.36	0.64

7. Prevalence and latent factor scores

Despite structural differences, women and men do not differ in the frequency of immune-related symptoms or in the mean *Immune* symptom score (all $p > .05$). These findings suggest that sex differences are more apparent in the structural organization of these symptoms than in overall symptom burden.

8. Impact of hormonal transitions on immune and autonomic regulation

Given the known impact of hormonal transitions on immune and autonomic regulation, we additionally explored whether the observed sex-specific factor structures differed between pre- and postmenopausal women. These analyses were considered exploratory due to limited subgroup sample sizes and because we could only roughly estimate which women were pre- and postmenopausal based on their age, since that information was not collected in the study.

Analyses stratified by illness duration revealed distinct patterns for immune- and gut-related symptoms. Across the total sample, flu-like symptoms did not differ significantly between duration categories. However, when stratified by menopausal status, a significant association emerged only in postmenopausal women, with flu-like symptoms decreasing as illness duration increased ($\chi^2(2)=7.01$, $p=.030$, $V=0.15$). A similar pattern was observed for susceptibility to infections, which also declined with longer illness duration in postmenopausal women ($\chi^2(2)=7.99$, $p=.018$, $V=0.16$), whereas no significant association was found in premenopausal women.

In contrast, gastrointestinal symptoms and food intolerances increased with illness duration in both groups. Gastrointestinal complaints showed a significant duration effect in premenopausal women ($\chi^2(2)=8.92$, $p=.012$, $V=0.15$) and a trend in postmenopausal women. Food intolerances demonstrated strong associations with illness duration in both groups (postmenopausal: $\chi^2(2)=17.72$, $p<.001$, $V=0.23$; premenopausal: $\chi^2(2)=8.73$, $p=.013$, $V=0.14$).

These findings indicate that immune-related symptoms are modulated by both menopausal status and disease stage, whereas gut-related symptoms show a progressive, duration-dependent increase independent of menopausal status (Table 7).

Table 7. Symptom prevalence across illness duration categories in pre- and postmenopausal women (*Duration of the disease*: ≤ 2 years, 3–8 years, > 8 years; *Menopausal status*: premenopausal: age ≤ 50 years, postmenopausal: age > 50 years)

Flu-like symptoms				
Group	$\chi^2(df=2)$	p-value	Cramer's V	Interpretation
Total	3.51	0.173	0.07	No correlation
Postmenopausal	7.01	0.030	0.15	Symptoms decrease with disease duration
Premenopausal	2.04	0.361	0.07	No correlation

Susceptibility to infections				
Group	$\chi^2(df=2)$	p-value	Cramer's V	Interpretation
Postmenopausal	7.99	0.018	0.16	Decrease over time
Premenopausal	5.11	0.078	0.11	Trend, not significant

Gastrointestinal complaints				
Group	$\chi^2(df=2)$	p-value	Cramer's V	Interpretation
Postmenopausal	4.40	.111	.12	Trend
Premenopausal	8.92	.012	.15	Increase over time

Food intolerances				
Group	$\chi^2(df=2)$	p-value	Cramer's V	Interpretation
Postmenopausal	17.72	< .001	.23	Significant increase over time
Premenopausal	8.73	.013	.14	Increase over time

Symptom burden

Across the full sample, women exhibited a significantly higher overall symptom burden compared to men. A linear regression model confirmed that the composite symptom score was higher in women ($\beta = -0.307$, $p = 0.010$), consistent with the group difference observed in the t-test ($t = 2.5907$). When examining the two symptom clusters separately, a clear pattern emerged: the *Gut* symptom score was significantly higher in women ($\beta = -0.226$, $p = 0.002$), whereas the *Immune* symptom score did not differ between sexes ($\beta = -0.080$, $p = 0.255$). Thus, the sex difference in overall symptom burden (*Combined Gut/Immune*) was driven exclusively by the *Gut* cluster.

Among women, menopausal status was associated with distinct symptom patterns. Postmenopausal women showed a significantly lower overall symptom burden than premenopausal women ($\beta = -0.236$, $p = 0.018$). This effect was specific to the *Immune* cluster, where postmenopausal women reported markedly fewer symptoms ($\beta = -0.177$, $p = 0.003$). In contrast, *Gut* symptoms did not differ by menopausal status ($\beta = -0.059$, $p = 0.358$). Interaction models including age (menopausal status $\times$ age) showed no evidence that age modified the association between menopausal status and either *Immune* or *Gut* symptoms (all $p > 0.65$). This indicates that the observed menopausal effect is not attributable to age differences, as the strength and direction of the association remained stable across the age range.

Because fewer women were represented at older ages, the corresponding estimates show larger standard errors and wider confidence intervals. Together, these results demonstrate that sex differences in symptom burden are cluster-specific and that menopausal status selectively influences *Immune*-related symptoms.

8. Summary of Sex-Specific Patterns

Taken together, the analyses reveal a consistent pattern of sex-specific differences in the organization of immune-related and gastrointestinal symptoms.

Symptom clusters across sexes. Flu-like symptoms and susceptibility to infections formed a coherent *Immune* cluster in both women and men, although associations were stronger in men, as reflected in higher tetrachoric correlations, higher factor loadings, and larger pseudo- R^2 values, pointing to a more tightly integrated immune symptom structure. Gastrointestinal complaints and (food) intolerances likewise formed a robust *Gut* cluster in both sexes, with comparable effect sizes but slightly higher bivariate correlations in men.

Combined Gut/Immune structure. When the four symptoms were modeled jointly, the analyses suggested a sex difference. In men, a one-factor solution fit the data very well ($\chi^2(2) = 3.50$, $p = .174$; RMSEA = .074; CFI = .981), consistent with a common immune–gastrointestinal dimension. In women, the same model fit poorly ($\chi^2(2) = 19.30$, $p = .0001$; RMSEA = .119; CFI = .913), indicating a separation into at least two distinct factors. Figure 1 summarizes these sex-specific factor structures.

Figure 2 summarizes the findings regarding the sex-specific pattern of gastrointestinal and immune-related symptoms.

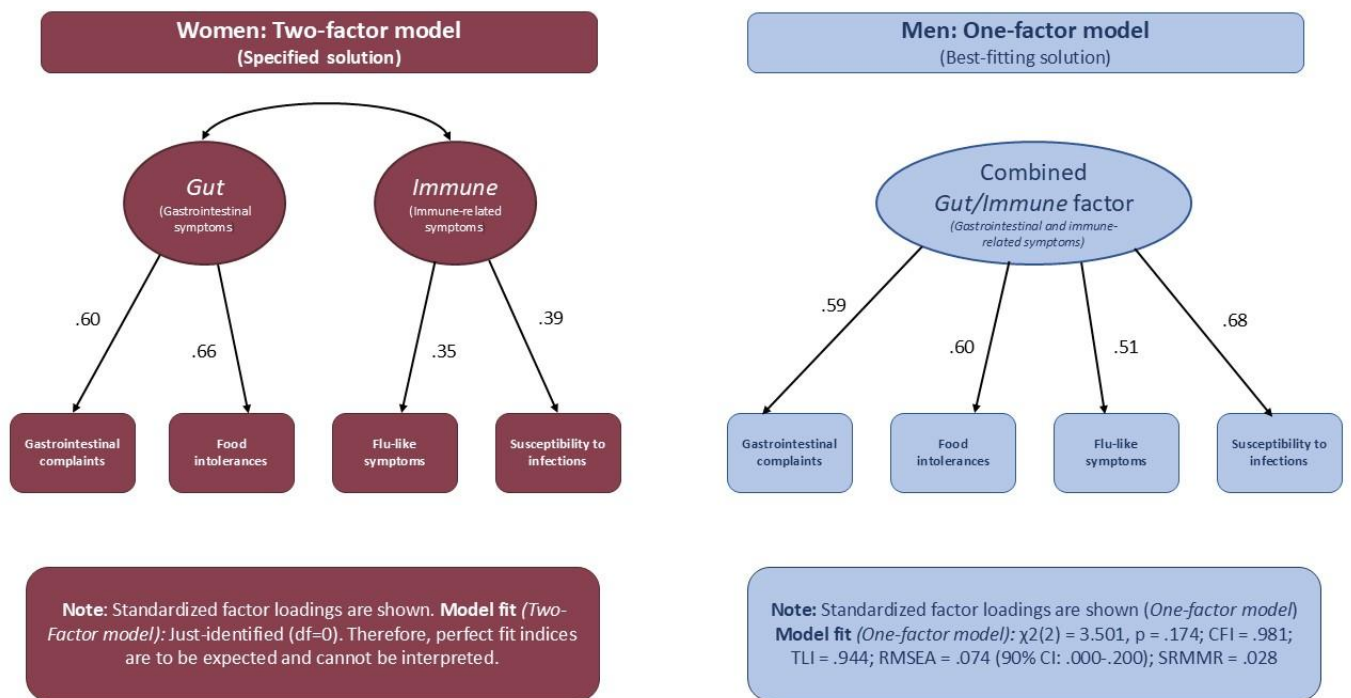


Figure 2 Sex-specific factor structures of immune-related and gastrointestinal symptoms.

Standardized factor loadings are shown. For women, a two-factor model provided the best representation of the data, separating immune-related symptoms (Susceptibility to infections, Flu-like symptoms) from gastrointestinal symptoms (Gastrointestinal complaints, Food intolerances). This model is just-identified ($df = 0$); therefore, global fit indices are not informative. For men, a one-factor model showed good fit ($\chi^2(2) = 3.50$, $p = .174$; CFI = .981; TLI = .944; RMSEA = .074, 90% CI: .000–.200; SRMR = .028), indicating that immune-related and gastrointestinal symptoms load on a single latent dimension.

Women. In women, *Immune* and *Gut* symptoms constituted two statistically distinct clusters with only modest cross-cluster correlations. Both logistic and linear models showed distributed predictive patterns, and factor analysis as well as SEM results were consistent with a two-domain conceptualization. Furthermore, women showed a higher overall symptom burden than men, driven exclusively by stronger *Gut*-related symptoms, while *Immune*-related symptoms did not differ between sexes.

Pre- versus postmenopausal women. The two-factor architecture remained stable across menopausal status, indicating that menopause does not alter the underlying symptom

organization. Flu-like symptoms were significantly more frequent in premenopausal women, consistent with the hypothesis that estrogen-related immune modulation may contribute to this. Premenopausal women showed persistently high *Immune* symptom levels independent of illness duration, potentially reflecting sustained immune-related symptom expression in the context of estrogen exposure. In contrast, *Immune*-related symptoms decreased with illness duration only in postmenopausal women, potentially consistent with a shift in symptom presentation across illness stages. Gastrointestinal symptoms and food intolerances increased with illness duration in both groups, possibly reflecting progressive alterations in gut-related symptomatology independent of menopausal status. Overall, *Immune* symptoms in women appear to be hormonally modulated, whereas *Gut* symptoms may represent a more stable, chronic symptom domain that progresses irrespective of the menopausal transition. Postmenopausal women reported fewer overall symptoms than premenopausal women, mainly due to lower *Immune*-related symptom burden. *Gut* symptoms were unaffected by menopausal status, and the effect was independent of age.

Men. In men, *Immune* and *Gut* symptoms formed a single integrated *Combined Gut/Immune* cluster with high cross-cluster correlations, and susceptibility to infections emerged as the dominant predictor across all models. Factor analysis and SEM consistently supported the one-factor structure. These findings should, however, be interpreted with caution given the smaller male subsample and the correspondingly wider confidence intervals. Therefore, replication in larger samples is required.

Discussion

This study provides robust evidence for sex-specific differences in the structure of immune related and gastrointestinal symptoms in ME/CFS. Across all statistical approaches—including tetrachoric correlations, logistic and linear regression, factor analysis, and structural equation modeling—women exhibited two distinct clusters (*Immune* and *Gut*), whereas men showed a single integrated cluster (*Combined Gut/Immune*) encompassing all four symptoms. Importantly, these structural differences do not coincide with differences in symptom prevalence or mean *Immune* factor scores. Symptom burden is comparable across sexes, but the latent organization of symptoms differs—it is not the frequency of symptoms but their inter symptom coupling that distinguishes women and men. These findings extend the results of Study 2, which validated the existence of *Immune* and *Gut* clusters in the overall sample [9],

and Study 3, which demonstrated sex-specific differences in neurocognitive sensory and autonomic symptom domains [11]. Together, these studies suggest that the symptom architecture of ME/CFS is not uniform but reflects sex-dependent neuroimmune–autonomic organization (Figure 2).

However, the proposed neuroimmune-autonomic mechanisms should be interpreted as biologically informed models derived from symptom architecture rather than direct mechanistic proof. Because microbiome composition, autonomic physiology, cytokine profiles, and intestinal permeability markers—to name but a few—were not directly measured, the present findings remain hypothesis-generating and require multimodal biological validation.

Sex-specific clustering patterns

- Women: Distinct Immune and Gut clusters

In women, immune related symptoms (flu-like symptoms, susceptibility to infections) and gastrointestinal symptoms (gastrointestinal complaints, food intolerances) formed two statistically separable clusters. Cross cluster correlations were modest, factor analysis yielded a clear two factor solution, and SEM confirmed a two-factor model fit the data substantially better than a one-factor model. Notably, when a unified one-factor *Combined Gut/Immune* model was tested in women, it showed poor fit, indicating the need for at least two latent dimensions. This structural divergence suggests that immunological and gastrointestinal symptoms are more differentiated in women and may be modulated by distinct hormonal, immunological, or neuro gastrointestinal mechanisms. These findings indicate that immune and gastrointestinal symptoms represent related but distinct biological axes in women.

This pattern aligns with known sex differences in immune function [19]. Women exhibit stronger adaptive immune responses, higher rates of autoimmune activation, and more pronounced cytokine reactivity than men [11]. They also show greater susceptibility to dysbiosis related gastrointestinal symptoms and higher prevalence of food intolerances, consistent with the stronger *Gut* cluster observed here. Study 1 similarly found that gastrointestinal symptoms and intolerances increased markedly during the early disease course in women, supporting the notion that these symptoms represent a distinct and clinically meaningful domain [8].

- Men: A Combined Gut/Immune Cluster

In contrast, men showed a markedly different pattern. All four symptoms loaded onto a single latent factor, with high tetrachoric correlations across domains and excellent SEM fit for a one-factor model. Across all analytic approaches, men consistently showed stronger associations than women: higher tetrachoric correlations, larger effect sizes in regression models, and higher factor loadings. The immunological symptom pattern (flu-like symptoms and susceptibility to infections) appeared particularly homogeneous, suggesting a more synchronized immunological response profile, and a similar—though slightly less pronounced—pattern emerged for gastrointestinal symptoms. Susceptibility to infections emerged as the dominant predictor in both logistic and linear regression models, suggesting that immune vulnerability may serve as a central organizing feature of symptom expression in men.

This *Combined Gut/Immune* cluster is consistent with biological evidence that men exhibit weaker humoral immunity, higher susceptibility to infections, and stronger sympathetic activation than women. Sympathetic overactivation can amplify inflammatory responses, impair gut barrier function, and promote gastrointestinal dysmotility—mechanisms that may couple immune and gut symptoms more tightly in men. Study 3 similarly found that men's autonomic symptom structure was dominated by respiratory and cardiovascular dysregulation, reflecting a *sympathetic–metabolic axis* rather than the *vagal–inflammatory axis* observed in women [10].

Mechanistic interpretation: Dysbiosis, intestinal permeability, and neuroimmune signaling

The sex-specific clustering patterns observed here may be interpreted within a mechanistic framework involving gut dysbiosis, intestinal barrier dysfunction (often referred to as ‘Leaky Gut’), and neuroimmune–autonomic signaling.

- *Dysbiosis and intestinal barrier dysfunction*

Multiple studies have documented reduced microbial diversity, loss of butyrate-producing species, and increased abundance of pro-inflammatory taxa in ME/CFS [20, 21]. Dysbiosis can impair epithelial tight junctions, increase intestinal permeability, and facilitate translocation of microbial products such as lipopolysaccharides (LPS). These processes can trigger systemic immune activation, oxidative stress, and cytokine release—mechanisms that plausibly contribute to both immune and gastrointestinal symptoms [22].

- *Sex-specific pathways*

The findings are consistent with a model in which women may exhibit a vagal–hormonal–inflammatory pathway: In women estrogen modulates and may enhance immune responses, epithelial barrier integrity, and vagal tone [23]. Dysbiosis-induced permeability may lead to LPS translocation and activation of the cholinergic anti-inflammatory reflex. This could contribute to why *Immune* and *Gut* symptoms remain distinct but interconnected in women [22].

The findings are consistent with a model in which men may exhibit a sympathetic–metabolic–vascular pathway: In men sympathetic dominance can impair gut motility, reduce mucosal perfusion, and promote inflammatory signaling [24, 25]. Vascular–metabolic stress may amplify immune vulnerability and gastrointestinal symptoms simultaneously [26]. This is compatible with the *Combined Gut/Immune* cluster observed in men.

These mechanisms are consistent with findings from Study 3, where women showed stronger vagal involvement in autonomic clusters, while men showed stronger sympathetic involvement [10].

- *The role of menopause*

In this study, we observed that flu-like symptoms were more prevalent among premenopausal women, whereas gastrointestinal symptoms and food intolerances did not differ between pre- and postmenopausal participants. These findings suggest that menopausal status modulates specific symptom domains but does not uniformly affect the broader symptom architecture. The pattern aligns with known immuno-endocrine mechanisms: estrogen modulates and may enhance antiviral and inflammatory immune responses [11, 18, 27-30], increase mast-cell reactivity and amplify central sensitivity to peripheral cytokine signals [31-33]. As a result, premenopausal women may exhibit a lower threshold for mounting a pronounced sickness response, which is characterized by fatigue, myalgia, and flu-like malaise [34]. After menopause, declining estrogen levels attenuate these immune-driven responses, which is compatible with the lower frequency of flu-like symptoms in older women.

Importantly, disease duration likely contributes to these differences. In our sample, postmenopausal women were older and had been ill for a longer period of time. Early disease stages in ME/CFS are typically marked by stronger antiviral and inflammatory signatures,

including interferon activity, cytokine peaks, and mast-cell activation—all of which drive flu-like symptoms [35-39].

One possible interpretation is that, over time, these acute immune features diminish in at least a subset of patients and are replaced by more chronic metabolic, autonomic, and vascular dysfunctions [40-42]. Thus, the combination of higher estrogen-dependent immune reactivity and shorter disease duration may predispose premenopausal women to more pronounced flu-like symptoms, whereas postmenopausal women may show attenuated immune-driven symptom expression due to both hormonal decline and longer disease chronicity.

In contrast, gastrointestinal symptoms and food intolerances were remarkably stable across menopausal status. This stability suggests that gut-related pathology may represent a more persistent and less hormonally responsive process. Current evidence indicates that once gut-barrier dysfunction is established—characterized by epithelial impairment, dysbiosis, and microbial translocation—it may become self-sustaining through ongoing immune activation and microbiome alterations. Menopause has been associated with long-lasting shifts in gut microbiome composition, reduced microbial diversity, and persistent markers of barrier dysfunction and microbial translocation [40, 43-46]. These findings support the possibility that gut dysfunction may persist beyond menopausal transition and may be less sensitive to hormonal modulation than immune-related symptoms. This may help explain why gastrointestinal symptoms did not differ between pre- and postmenopausal women in our cohort despite clear hormonal modulation of immune-related symptoms.

Our additional analyses revealed that disease duration modulated immune-related symptoms only in postmenopausal women. In this group, both flu-like symptoms and susceptibility to infections decreased with longer illness duration, whereas no such pattern was observed in premenopausal women. This potentially aligns with the previously described notion that immune-driven symptoms are strongest in early disease stages and attenuate over time, particularly once estrogen-dependent immune amplification is lost. In contrast, gastrointestinal symptoms and food intolerances increased with disease duration in both groups, indicating a progressive and hormonally independent process consistent with persistent gut-barrier dysfunction. Together, these findings reinforce a dual-process model in which immune-related symptoms are shaped by both hormonal status and disease stage, whereas *Gut*-related symptoms reflect a stable, chronic pathology that progresses irrespective of menopausal transition.

Taken together, our findings suggest a dual-process model: immune-related symptoms are dynamic and hormonally modulated, whereas *Gut*-related symptoms reflect a stable, structural pathology that persists independently of menopausal status. This distinction may have implications for personalized treatment strategies, highlighting the need to consider both hormonal context and disease stage when interpreting symptom profiles in women with ME/CFS.

The present findings also align with the broader symptom architecture identified in Studies 2 and 3 [9, 10].

- Women: A multi-axis, interconnected system

Women show distinct *Immune* and *Gut* clusters, a coherent *Autonomic* cluster, a homogeneous *Brain/Brain2* cluster, and moderate correlations across clusters.

This pattern suggests a vagal–hormonal–inflammatory network, in which immune, gastrointestinal, autonomic, and neurocognitive symptoms are interconnected but retain distinct latent structures. Women exhibit a pattern of partially segregated but interconnected symptom clusters, consistent with recent systems-immunology and neuroendocrine research. Contemporary work shows that women display more coherent yet distinctly modular immune and gut signatures, shaped by estrogen-dependent transcriptional programs and X-linked immune regulation [23, 47]. At the same time, multimodal analyses are consistent with moderately correlated but structurally separable immune, metabolic, autonomic, and neurocognitive modules, reflecting sex-specific integration across inflammatory, hormonal, and neurovisceral axes [48]. Neurocognitive domains in women also tend to form more homogeneous clusters, consistent with hormone-sensitive neuroimmune coupling [49]. Together, these findings support the interpretation of a vagal–hormonal–inflammatory network in which immune, gastrointestinal, autonomic, and brain-related symptoms remain distinct in their latent structure while being functionally linked through shared regulatory pathways.

- Men: A more integrated, sympathetic-driven system

Men show a *Combined Gut/Immune* cluster, an *Autonomic* cluster dominated by respiratory and cardiovascular symptoms, and a more heterogeneous *Brain/Brain2* cluster.

This pattern suggests a sympathetic–metabolic–vascular network, in which immune and gastrointestinal symptoms are tightly coupled and interact strongly with autonomic

dysregulation. Recent neuroimmune and autonomic research supports the interpretation that men exhibit a more integrated sympathetic–metabolic–vascular network, consistent with a unified *Combined Gut/Immune* cluster and stronger coupling between immune and gastrointestinal symptoms. Contemporary work demonstrates that sympathetic activation tightly links metabolic stress, vascular tone, and inflammatory signaling, producing coherent immune–gastrointestinal modules that interact strongly with autonomic dysregulation [50, 51]. Parallel findings in gastrointestinal neuroimmune disorders show that immune and gut pathways become increasingly synchronized under sympathetic dominance, reinforcing the emergence of a single immune–gastrointestinal cluster [52]. Moreover, studies of the gut–immune–brain axis indicate that men display more heterogeneous neurocognitive patterns alongside stronger immune–autonomic coupling, aligning with a more variable *Brain/Brain2* cluster and a distinct *autonomic* cluster dominated by respiratory and cardiovascular symptoms [53]. Together, these findings support a model in which immune, gastrointestinal, autonomic, and neurocognitive symptoms in men are organized by a sympathetic–metabolic–vascular architecture, producing tighter interconnections than those observed in women (Table 1). The exploratory factor analysis further reinforces these sex-specific patterns. In women, as described above, immune and gastrointestinal symptoms separated cleanly into two latent factors, with moderate loadings for immune symptoms and stronger, more coherent loadings for *Gut* symptoms. This structure indicates that immune and gastrointestinal domains represent distinct but interconnected biological axes, consistent with a modular, vagal–hormonal–inflammatory architecture. In men, by contrast, all four symptoms loaded onto a single integrated factor, reflecting a tightly coupled immune–gastrointestinal dimension. The high cross domain correlations and unified factor structure support the interpretation that men exhibit a sympathetic–metabolic–vascular pathway, in which immune vulnerability, gastrointestinal dysregulation, and autonomic activation converge into one latent construct. Together, these EFA findings align with the broader SEM results and underscore that the organization of *Immune* and *Gut* symptoms in ME/CFS is fundamentally sex-dependent.

Clinical and methodological implications

Taken together, these results underscore the importance of sex-specific modeling in ME/CFS research (Figure 3). Because symptom prevalence and mean factor scores are comparable between sexes while the latent symptom architecture differs fundamentally, analyses that pool men and women risk obscuring clinically relevant structural patterns. The findings therefore provide a foundation for more nuanced diagnostic approaches and sex-tailored therapeutic

strategies that target the distinct neuroimmune–autonomic configurations observed in women and men.

Clinical and translational implications

These findings have several implications:

(a) Sex-specific diagnostic patterns

Pre- and postmenopausal women may benefit from separate assessment of immune and gastrointestinal symptoms. In men, these symptoms may represent a single underlying process.

Sex-specific differences in symptomatology and immune function suggest that tailored diagnostic assessment may increase diagnostic precision. In cardiology, sex-specific high-sensitivity troponin thresholds have been shown to reduce underdiagnosis of myocardial infarction in women and to identify high-risk patients who would otherwise be missed [54-56]. Up to now, sex-specific differences in symptom patterns, immune and hormonal profiles in ME/CFS are increasingly well documented [6, 57-59], but prospective studies demonstrating a direct clinical benefit of sex-stratified diagnostic pathways are still lacking.

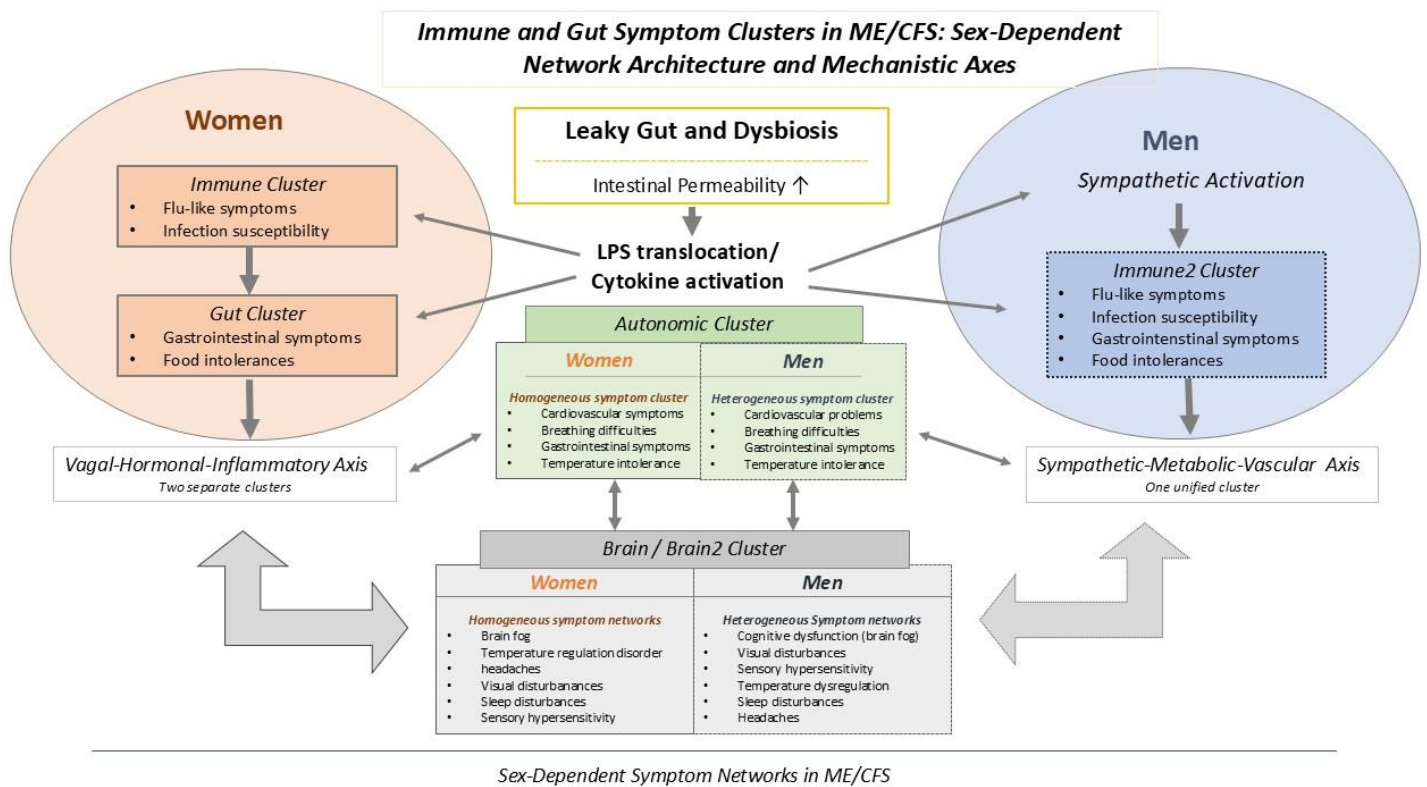


Figure 3 *Mechanistic model of sex-specific neuroimmune–autonomic pathways in ME/CFS*

Illustration summarizing hypothesized biological mechanisms linking intestinal dysbiosis and barrier dysfunction (often referred to as ‘Leaky Gut’) to immune activation and neuroimmune–autonomic symptom clustering in myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS). The central panel (top) depicts increased intestinal permeability and microbial translocation (lipopolysaccharides [LPS], cytokines) as initiating events. Left pathway (women): LPS induced cytokine release activates vagal afferents and estrogen modulated immune signaling, producing distinct *Immune* and *Gut* clusters mediated by a vagal–hormonal–inflammatory axis. Right pathway (men): the same inflammatory stimuli amplify sympathetic activity, vascular stress, and metabolic dysregulation, generating a *Combined Gut/Immune* cluster through a sympathetic–metabolic–vascular axis. Both pathways converge on downstream *Brain/Brain2* (neurocognitive) and *Autonomic* clusters, illustrating how peripheral immune–gut processes interact with central regulation to form sex-dependent symptom networks in ME/CFS (see middle panel below).

b) *Potential biomarkers*

The sex-specific symptom architectures identified in this study may also have implications for biomarker research in ME/CFS. The distinct separation of immune and gastrointestinal symptom clusters in women, compared with the integrated *Combined Gut/Immune* cluster observed in men, suggests that different biological pathways may underlie symptom organization across sexes and may therefore require sex-stratified biomarker approaches.

In women, biomarkers related to intestinal barrier dysfunction, microbiome composition, and vagal-immune regulation may be particularly relevant. Potential candidates include short-chain fatty acid (SCFA) profiles, markers of microbial translocation and intestinal permeability such as zonulin, lipopolysaccharide-binding protein (LBP), and soluble CD14 (sCD14), as well as estrogen-sensitive cytokine signatures and mast-cell-associated mediators. Because the female symptom architecture was characterized by partially distinct but interconnected immune and gastrointestinal domains, autonomic measures reflecting vagal regulation—particularly heart rate variability (HRV)—may also provide useful indicators of neuroimmune integration.

In men, the tightly coupled *Combined Gut/Immune* cluster and stronger association with autonomic dysregulation suggest that biomarkers related to sympathetic activation, vascular stress, and metabolic dysfunction may be more informative. Potential candidates include catecholamine-associated autonomic markers, sympathetic HRV indices, endothelial dysfunction markers, inflammatory-metabolic stress markers, and immune profiles associated with infection susceptibility or chronic low-grade inflammation. The stronger convergence of immune and gastrointestinal symptoms in men may reflect a more integrated neuroimmune-metabolic response pattern. Altered natural killer cell function and immune dysregulation have repeatedly been described in ME/CFS and may contribute to the immune-related symptom architecture observed here [60].

Importantly, these proposed biomarkers should be considered hypothesis-generating rather than disease-specific indicators. Because the present study did not directly assess microbiome composition, autonomic physiology, cytokine profiles, or intestinal permeability, the suggested biomarker pathways remain inferential and require validation in multimodal longitudinal studies. Future research integrating symptom-network analysis with immune, autonomic, microbiome, metabolic, and vascular biomarkers may help identify biologically meaningful sex-specific endophenotypes in ME/CFS.

(c) Therapeutic strategies

The present findings suggest that sex-specific symptom architectures in ME/CFS may also have implications for the development of more individualized therapeutic approaches. Because women and men showed fundamentally different patterns of immune–gastrointestinal coupling, therapeutic strategies targeting neuroimmune, autonomic, and gastrointestinal pathways may need to be adapted according to sex-specific biological organization.

In women, the separation of *Immune* and *Gut* clusters together with the broader vagal–hormonal–inflammatory pattern may indicate that interventions targeting gut-barrier integrity, microbiome composition, and neuroimmune regulation could be particularly relevant. Potential approaches may include microbiota-targeted interventions, dietary modulation, anti-inflammatory therapies, mast-cell–stabilizing strategies, and treatments aimed at improving vagal regulation, such as breathing-based autonomic interventions or vagal nerve stimulation. Because immune-related symptoms in women are consistent with a model in which these are influenced by menopausal status and disease duration, hormonal context and disease stage may be important considerations when evaluating therapeutic responses.

In men, the integrated *Combined Gut/Immune* cluster and stronger association with sympathetic–metabolic–vascular dysregulation suggest that therapies targeting autonomic imbalance, vascular function, and metabolic stress may warrant further investigation. Potential approaches could include autonomic regulation strategies, breathing and pacing interventions, treatments aimed at reducing sympathetic overactivation, vascular-supportive therapies, and infection-related or immunomodulatory interventions in selected patients. The tighter coupling between immune and gastrointestinal symptoms observed in men may indicate that combined neuroimmune–metabolic interventions are particularly relevant in this subgroup.

Importantly, these proposed therapeutic implications remain exploratory and should not be interpreted as evidence-based treatment recommendations. The present study did not directly assess autonomic physiology, microbiome composition, inflammatory biomarkers, or treatment responses. Consequently, the suggested therapeutic pathways are derived from symptom-network architecture and mechanistic interpretation rather than direct interventional evidence. Nevertheless, the findings support the broader concept that ME/CFS may comprise biologically distinct, sex-dependent neuroimmune–autonomic phenotypes that could benefit from more stratified and personalized treatment approaches.

Future prospective studies combining symptom-network analysis with biological profiling and longitudinal treatment outcomes may help determine whether sex-specific therapeutic strategies improve clinical response, symptom burden, or disease progression in ME/CFS.

Limitations

Several limitations should be considered when interpreting these findings. First, menopausal status was operationalized using an age-based cutoff (<50 vs. ≥ 50 years), which provides only a proxy for biological menopausal transition and does not capture perimenopausal fluctuations, hormonal variability, or the potential influence of hormone replacement therapy. Because detailed menopausal information was not collected, classification into pre- and postmenopausal groups remained approximate.

Second, the analyses relied on cross-sectional data, precluding conclusions about temporal dynamics or causal relationships between hormonal status and symptom expression. The identified latent variables therefore represent statistical symptom structures rather than directly observed biological mechanisms.

Third, all symptoms were assessed via self-report and dichotomized prior to analysis. Dichotomization reduces variance and may attenuate factor loadings, while self-report measures are additionally susceptible to recall and reporting biases. Because each latent domain was represented by only two indicators and the overall model included four binary items, the factor analyses should be considered exploratory and hypothesis-driven. Accordingly, the factor loadings are interpreted primarily as indicators of whether the observed symptom patterns are consistent with the proposed latent structure, rather than as precise estimates of underlying biological processes.

Fourth, confirmatory factor analyses (CFA) and SEM were estimated using maximum likelihood (ML) in Stata 12.0 because categorical estimators such as WLSMV were not available in the software version used. Given the binary nature of the indicators, ML estimation may produce biased standard errors and fit indices. However, supplementary analyses based on tetrachoric correlations and cross-validation suggested that the latent structure was reasonably robust.

Fifth, subgroup sample sizes, while adequate for the primary analyses, limited statistical power to detect subtle structural differences between pre- and postmenopausal women and prevented formal tests of measurement invariance across groups.

Finally, no biological markers were collected. Consequently, proposed mechanisms involving autonomic regulation, microbiome alterations, inflammatory pathways, or related physiological processes remain inferential and should be regarded as hypothesis-generating. In addition, the analyses focused on a relatively restricted set of immune-related and gastrointestinal symptoms; broader symptom domains such as autonomic, neurological, or musculoskeletal symptoms may reveal additional menopausal effects not captured in the present study.

Conclusions

Immune and gastrointestinal symptoms in ME/CFS demonstrate fundamentally different clustering patterns in women and men. Women exhibit two partially separable symptom clusters, whereas men show a more integrated symptom architecture. These sex-specific patterns may reflect divergent neuroimmune–autonomic mechanisms and support the assumption that biological sex substantially shapes symptom organization in ME/CFS. Accordingly, the present findings provide further support for the development of sex-sensitive diagnostic frameworks, mechanistic models, and therapeutic approaches. In combination with Studies 1–3, the current results contribute to an emerging translational model of ME/CFS as a sex-modulated neuroimmune–autonomic disorder characterized by distinct immunological and gastrointestinal symptom trajectories.

Furthermore, our findings indicate that menopausal status selectively modulates immune-related symptom expression, whereas gastrointestinal symptoms remain remarkably stable across pre- and postmenopausal women. Flu-like symptoms were more prevalent in premenopausal participants, consistent with evidence for estrogen-dependent amplification of antiviral and pro-inflammatory immune pathways as well as the heightened neuroimmune reactivity typically associated with earlier or more active disease stages. By contrast, food intolerances and related gastrointestinal symptom patterns did not differ significantly between pre- and postmenopausal groups. This observation may support the hypothesis that gastrointestinal manifestations in ME/CFS are linked to persistent alterations of intestinal

barrier integrity and gut–immune interactions that, once established, become relatively chronic and less dependent on hormonal status.

Taken together, these findings support a dual-process model in which immune-related symptoms appear comparatively dynamic and hormonally modulated, whereas gastrointestinal symptoms may represent a more stable and self-perpetuating pathophysiological process. The apparent dissociation between immune and gastrointestinal symptom trajectories further suggests that partially distinct biological mechanisms contribute to symptom persistence in ME/CFS. Recognizing these differential trajectories may improve clinical phenotyping, facilitate stratification in future research, and support the development of more targeted and individualized therapeutic interventions, particularly for women across different hormonal stages.

Several limitations should nevertheless be considered. The cross-sectional design precludes causal inference regarding the temporal relationship between hormonal changes and symptom development. In addition, menopausal status was assessed categorically rather than through direct hormonal measurements, and symptom data relied on self-report measures. Future longitudinal studies integrating endocrine, immunological, autonomic, and microbiome-related biomarkers will be important to further clarify the mechanistic interplay between sex hormones, neuroimmune regulation, and gastrointestinal dysfunction in ME/CFS.

Acknowledgements

The authors would like to thank all study participants who gave us their time to participate in the study. Many thanks also to the ME/CFS patient organizations and the two ME/CFS centers in Berlin and Munich for forwarding the call for participation in the study and to the state parliament of Baden-Württemberg for partially funding the study (AZ: M57-5434-9/2).

Author's Contributions

- (1) LHH contributed to the development and conception of the research project, the preparation, collection, procurement, and provision of the data, the analysis and interpretation of the data, the review of the sources, the conclusions drawn from these findings, and the writing of the manuscript.
- (2) LMH contributed to the preparation, provision, and interpretation of the data.

Funding

The survey of ME/CFS patients underlying this publication was partially funded by the Ministry of Social Affairs, Health and Integration using state funds approved by the Baden-Württemberg State Parliament (file number: M57-5434-9/2). All subsequent studies based on this survey, including the present study, were not funded.

Data availability

All data supporting the results of this study are included in the article or can be requested from the corresponding author.

Declarations

Ethic approval

This study was conducted in 2022 in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Furtwangen University, Germany (application number: 22-057).

Consent for publication

Not applicable

Consent to participate & written Consent for publication

All participants were fully informed about the objectives and procedure before the study began and gave their written consent to participate. See *Information and declaration of consent* in the "Supplementary files" section

Code availability

Not applicable.

Additional Notes

The translation from German was supported by www.DeepL.com/Translator (free version). Furthermore, we would like to thank the AI tool ChatGPT (OpenAI) for language editing and wording optimization in individual sections under human guidance. All ideas, data and interpretations originate from the authors.

Conflicts of interest

The corresponding author declares on behalf of all authors that there is no conflict of interest.

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